



National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – M.Sc. Digital Forensics and Information Security		Sem – II
Date –03/02/26		
Subject Name – Artificial Intelligence		Subject Code – CTMSDFIS SII P4
Time – 45 minutes		Max. Marks– 25
I	Short Answer Questions	15 marks
I.	Differentiate between <code>append()</code> and <code>extend()</code> methods in Python lists with a brief example.	2 marks
II.	Write a Python command to create a dictionary named <code>student</code> with keys "name" and "age", and then add a new key "grade" to it.	2 marks
III.	Define a user-defined function named <code>calculate_area</code> that accepts <code>radius</code> as an argument and returns the area of a circle.	2 marks
IV.	A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?	2 marks
V.	Write a <code>while</code> loop that prints the even numbers from 2 to 10 inclusive.	2 marks
VI.	Explain the difference between a Python list and a NumPy array. Why is NumPy preferred for Data Science?	2.5 marks
VII.	Calculate the Mean and Median of the following dataset: [4, 8, 2, 10, 4, 5].	2.5 marks
II	Answer any 2 questions (2 x 5 marks each)	10 Marks
i.	Write a short note on Linear Regression.	5 marks
ii.	Explain Correlation. Compute Pearson Correlation Coefficient given $X = [2, 4, 6, 8]$ and $Y = [3, 5, 7, 10]$	5 marks
iii.	Multiply the following matrices (dot product): $A \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$ and $B \begin{bmatrix} 1 & 5 \\ 2 & 0 \end{bmatrix}$	5 marks

All the best